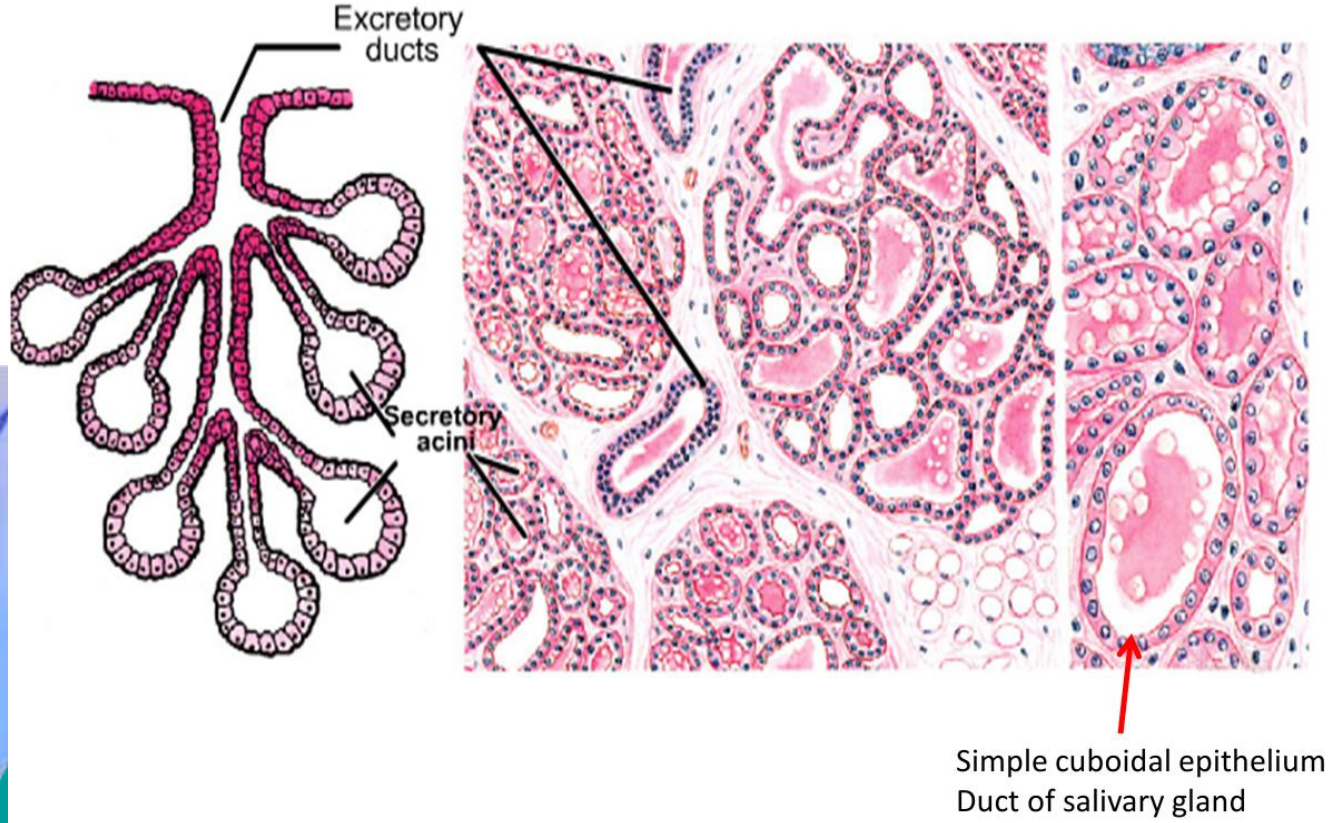
Histology – Salivary glands

Dr. Madhan



Salivary gland



GLANDS - TYPES

- Depending on the type of secretions
- Mucous glands
  - (mucopolysaccharide, seen to collect in the apex of the cell so the nucleus is pushed to the base)

- Serous glands
  - (protein in nature, cytoplasm granular, nucleus is central)

GLANDS - TYPES

Based on how the secretions are poured out of the cell:
MEROCRINE -secretions are thrown by exocytosis, cell remaining intact. eg: salivary glands, pancreas

APOCRINE - part of the cell is lost along with its secretions eg. Sweat gland

HOLOCRINE - Entire cell degenerates while discharging its secretions. eg. Sebaceous gland.

Endocrine glands:

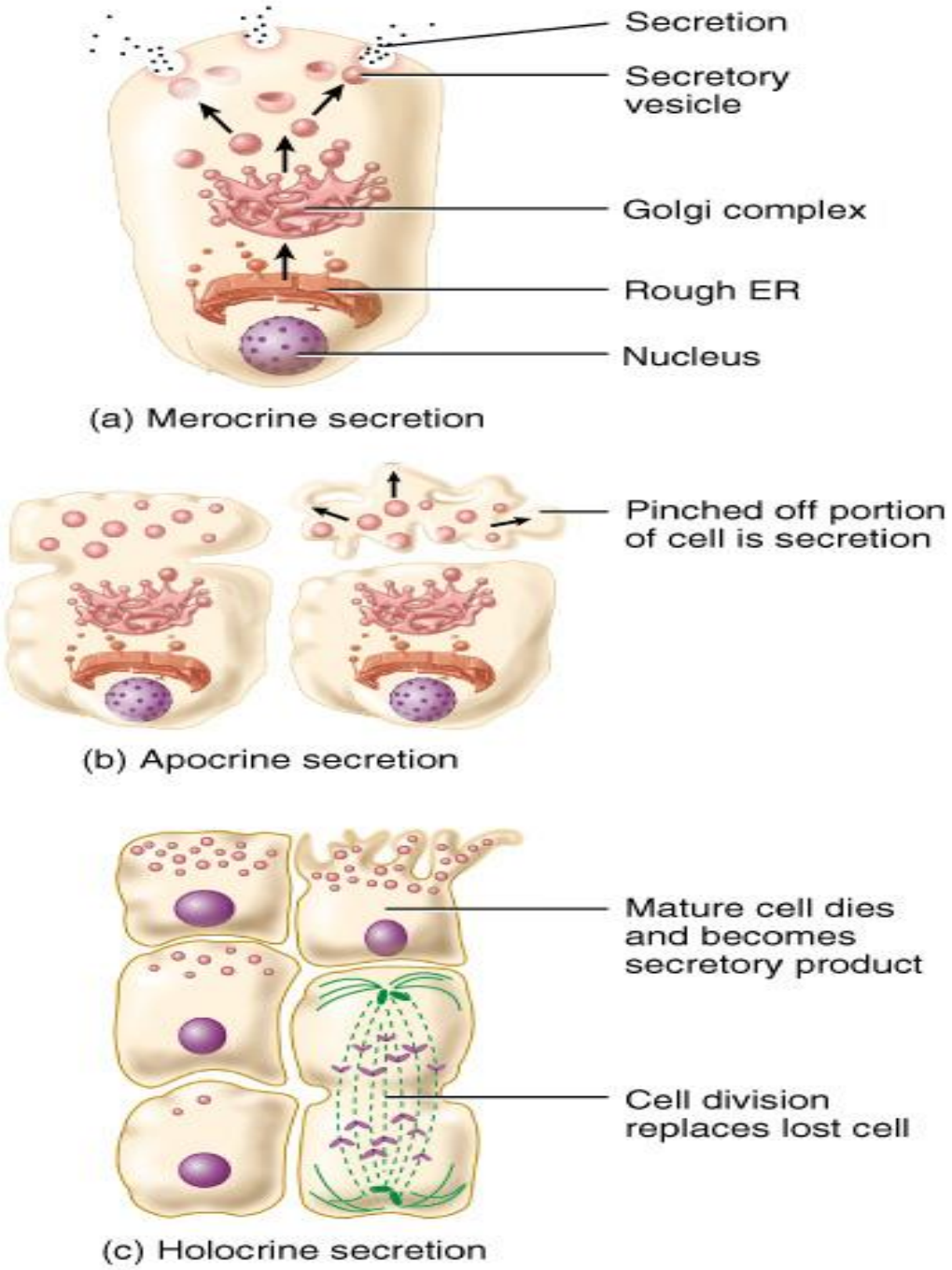
Arranged in clumps
Has good blood supply
Supported by connective tissue, to hold the clumps of cells and blood vessels in place
Surrounded by capsule.

GLANDS - TYPES

MEROCRINE GLANDS

APOCRINE GLANDS

HOLOCRINE GLANDS



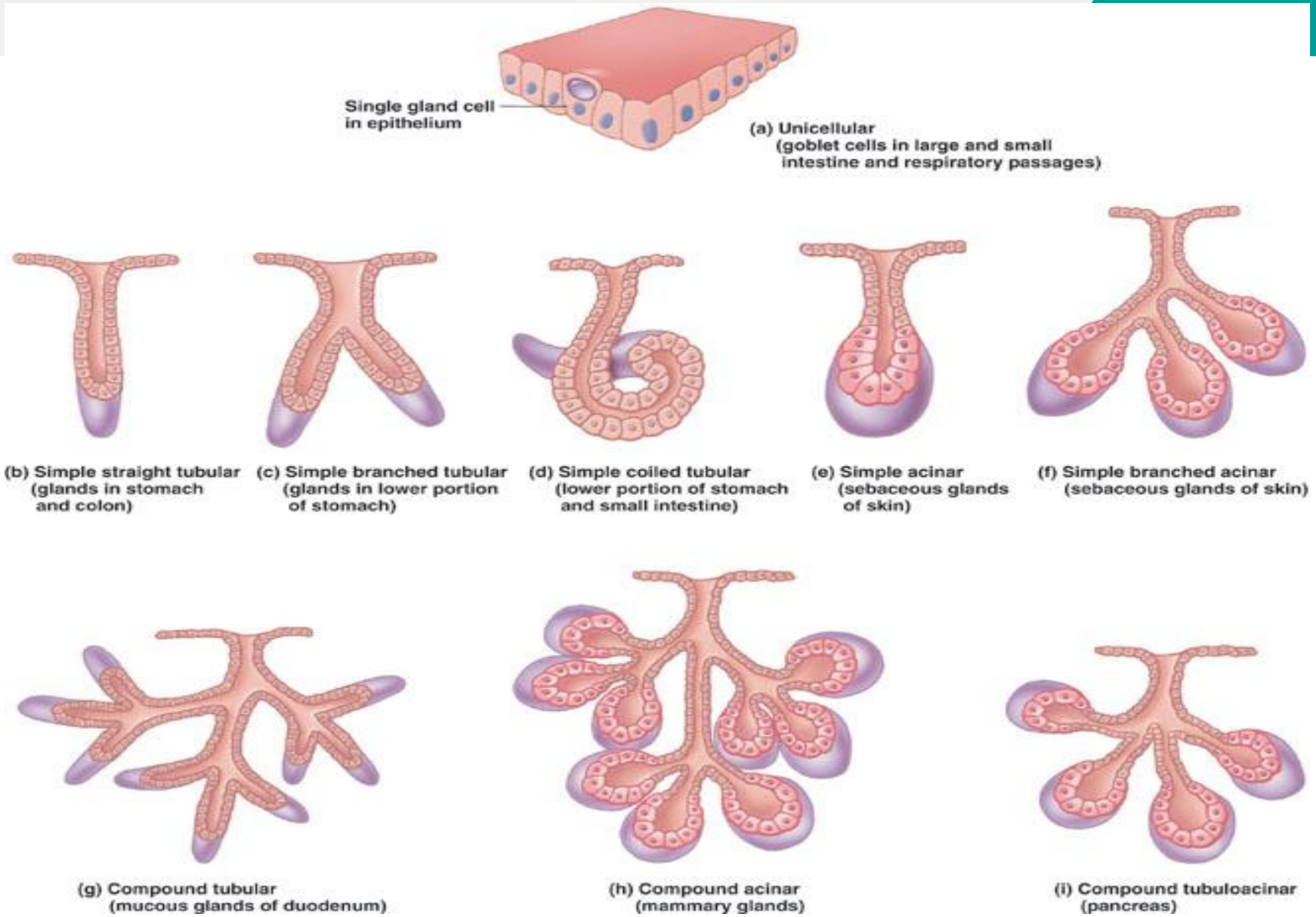
GLANDS - TYPES

- EXOCRINE GLAND are classified based on the ducts and their branching

- SIMPLE GLAND
  - If the secretions are sent through one duct

- COMPOUND GLAND
  - Sometimes number of secreting glands, each group with their own duct, these ducts unite to form larger ducts which ultimately drain to the epithelial surface (via these larger ducts)

GLANDS – Types based on their shape



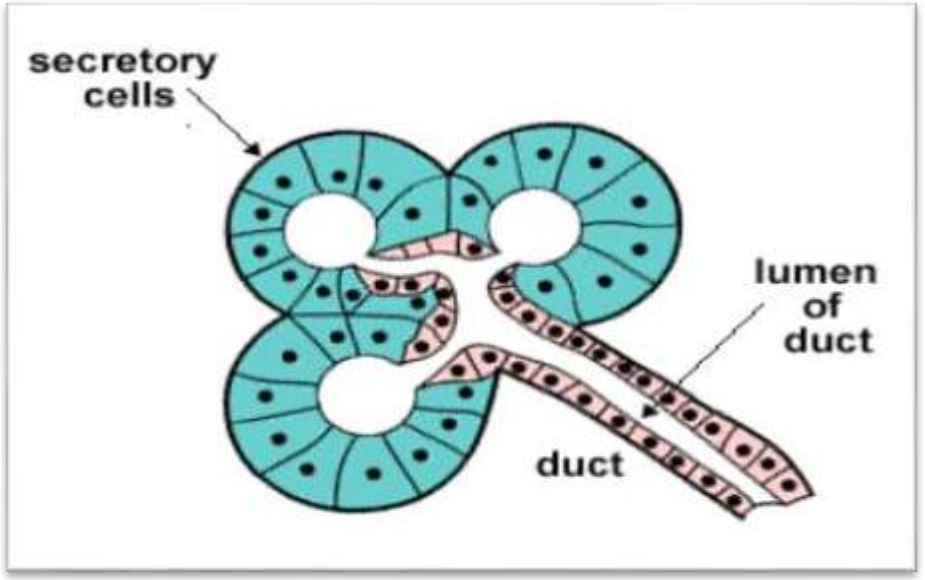
Exocrine gland

Has a secretory unit, ducts and connective tissue that support these components

1. **Secretory unit**
   - Secretory epithelial cells
   - Released their secretion into a lumen

2. **Duct**
   - A tube lined by epithelium
   - Conveys secretion from a secretory unit to the surface





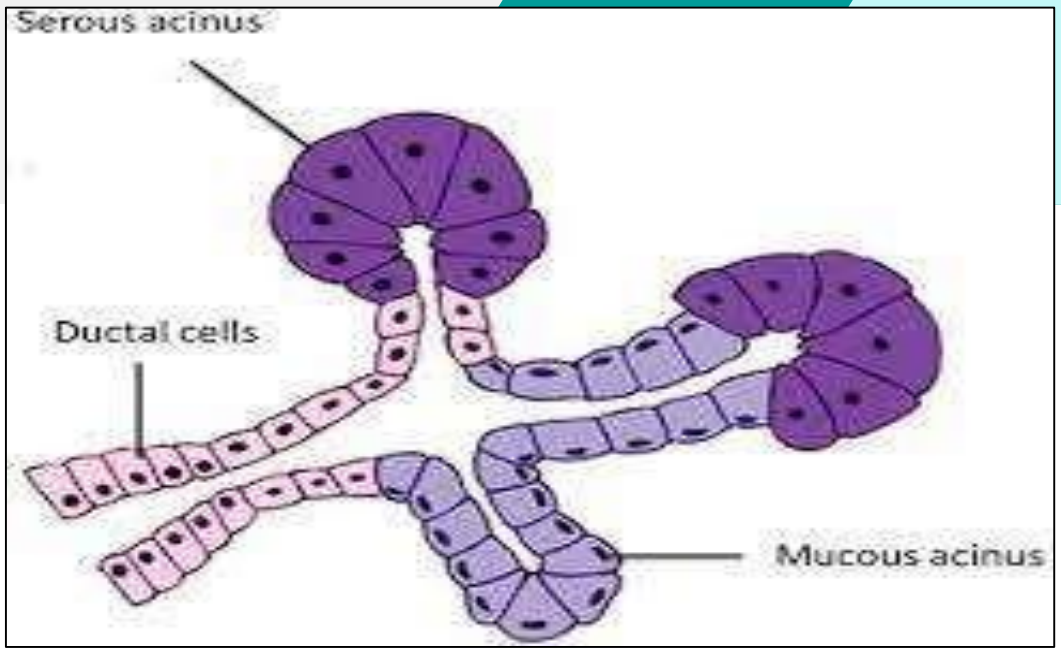
Secretory unit

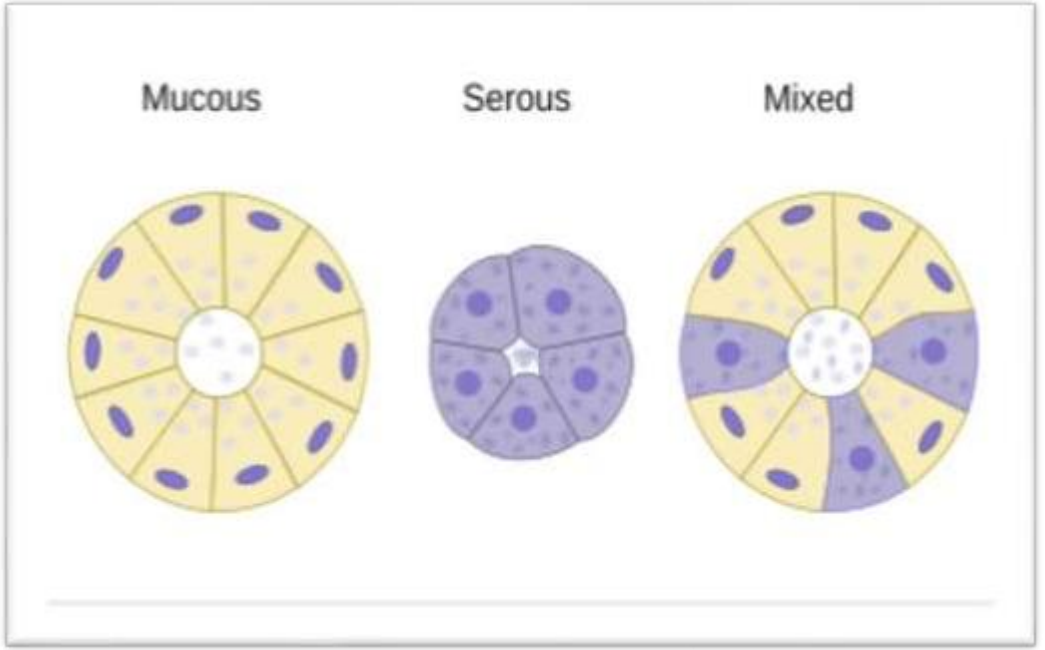
- Salivary glands contain the secretory units which are called acini.

- Classified according to the cell types they contain:

1. **Serous acini**
- Contain only serous cells
- Generally, are pyramidal in shape with spherical or round basal nucleus

2. **Mucous acini**
- Contain only mucous cells
- Generally are tubular,flattened nucleus at the base

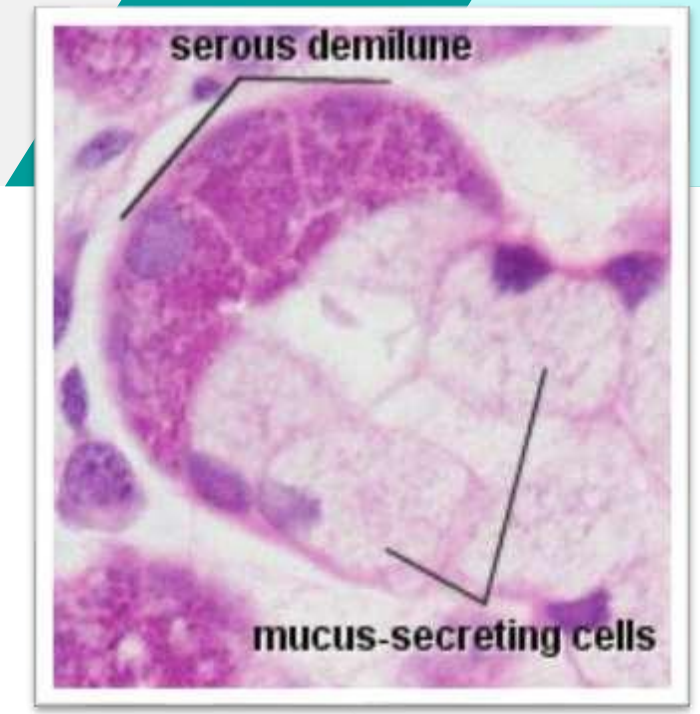


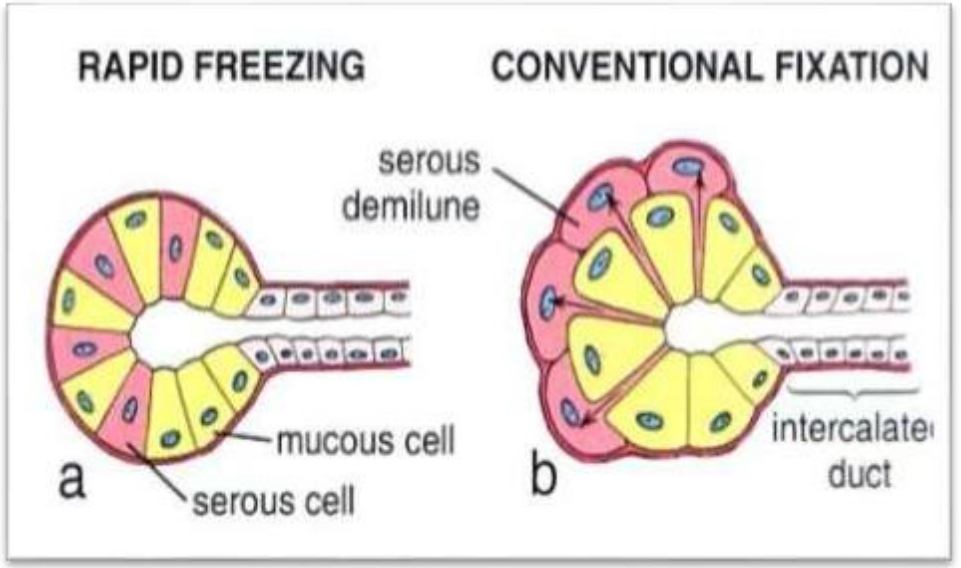


3. **Mixed (or seromucous) acini**

- A mixture of serous and mucous cells.

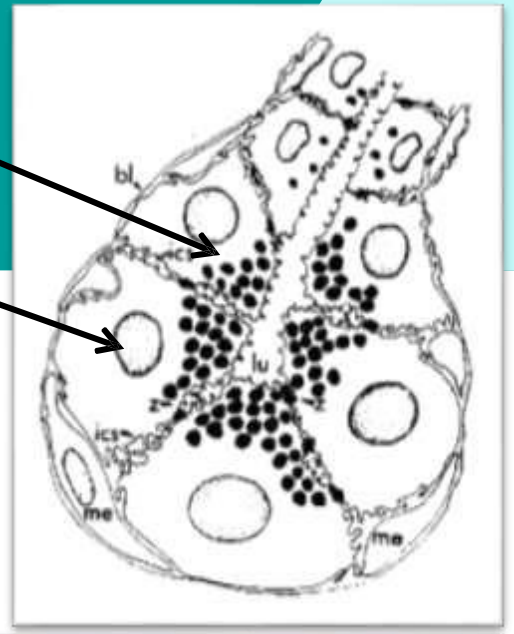
- In routinely prepared tissue sections, mucous cells swell and push serous cells into a peripheral cap known as a **serous demilune.**

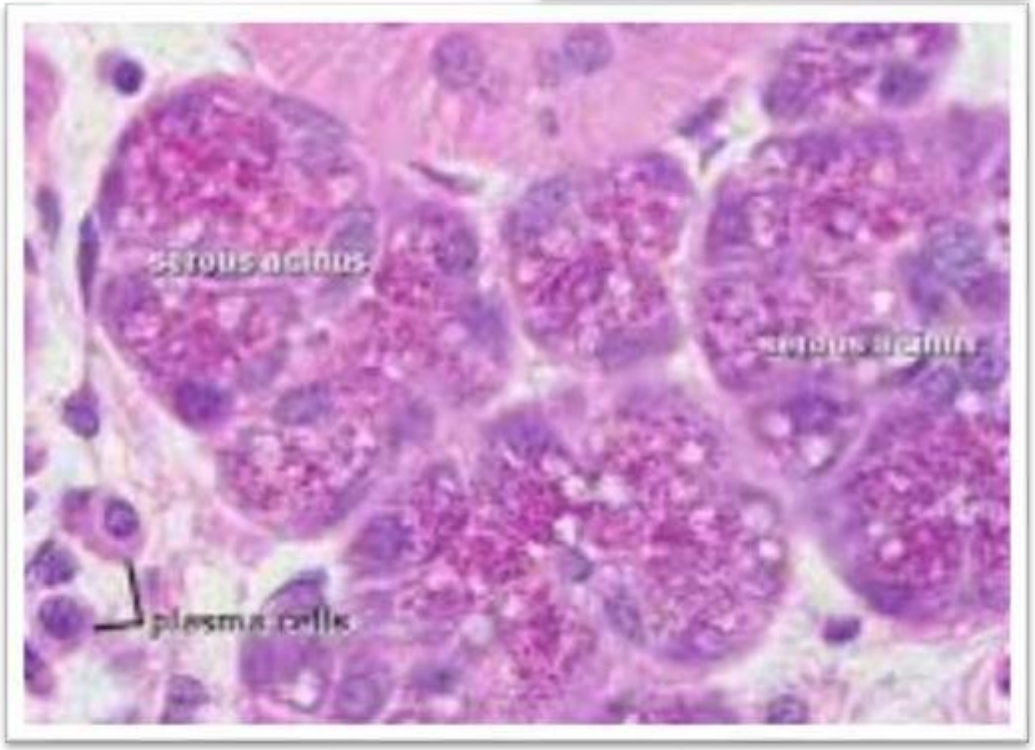




1. Serous cells

- Pyramid shaped cells
  - ✓ Base of cell rests on basal lamina
  - ✓ Apex facing lumen of acinus
- Nucleus is round, located at the base.
- Contain numerous rER, free ribosomes & numerous secretory granules.
- Serous secretions are watery.
- Secretes non-glycosylated protein secretions e.g. amylase

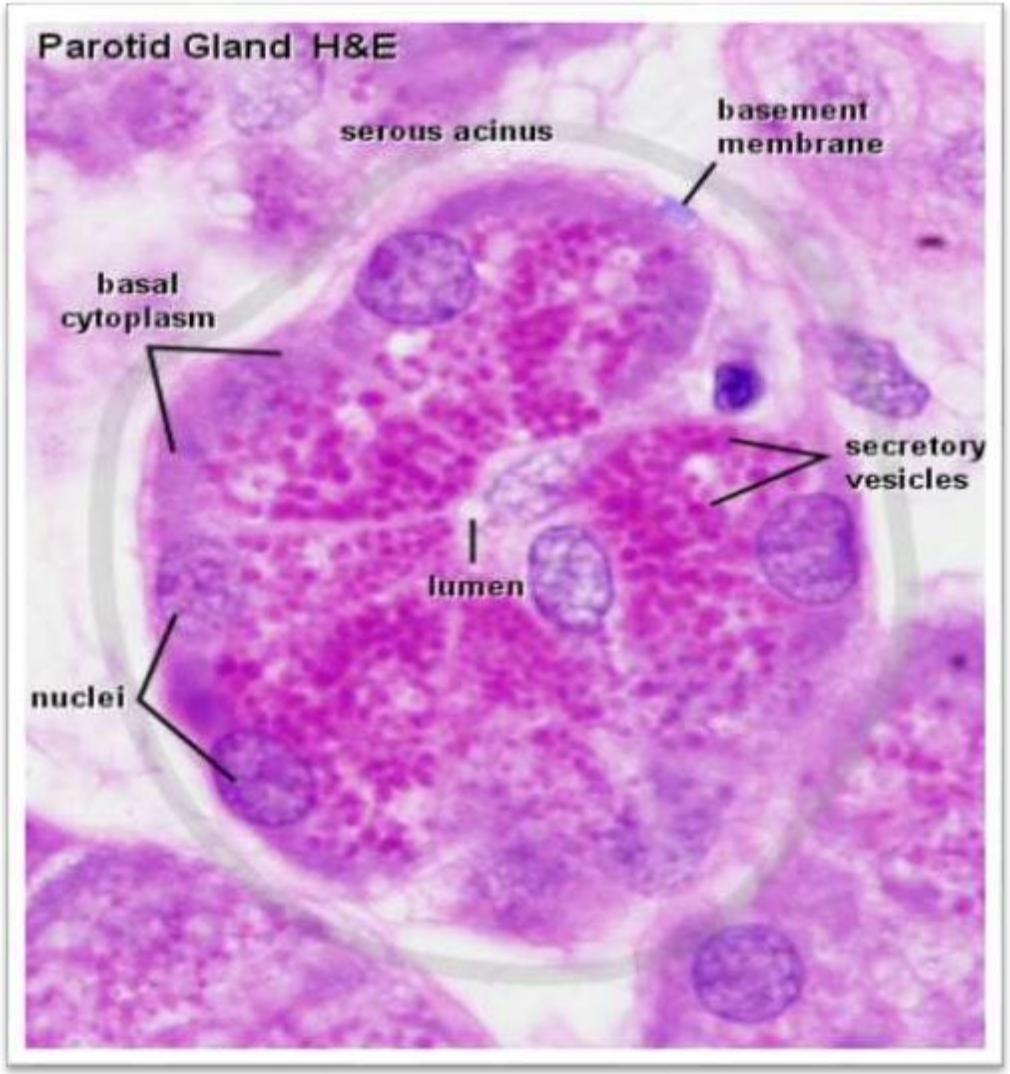




In H&E sections:

- Basal cytoplasm stains basophilic due to presence of numerous rER & ribosomes.

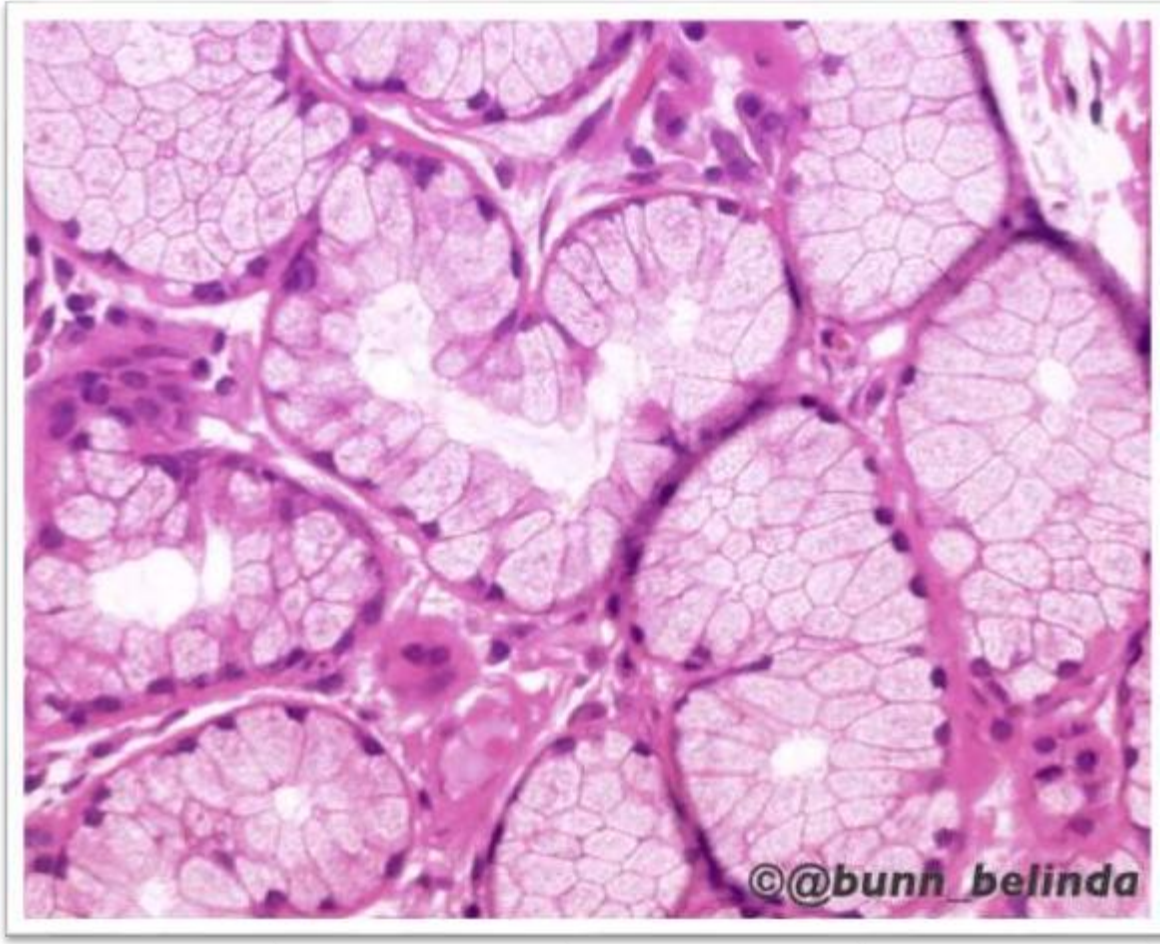
- Apical cytoplasm stains acidophilic due to presence of secretory granules. (if well preserved)



2. Mucous cells

- Cell flattened due to secretory products at apical cytoplasm
- Flat nucleus located at the base
- Secretory products contain mucinogen granules
- Secretes mucous which is viscous in nature (glycosylated protein) e.g. sialomucins



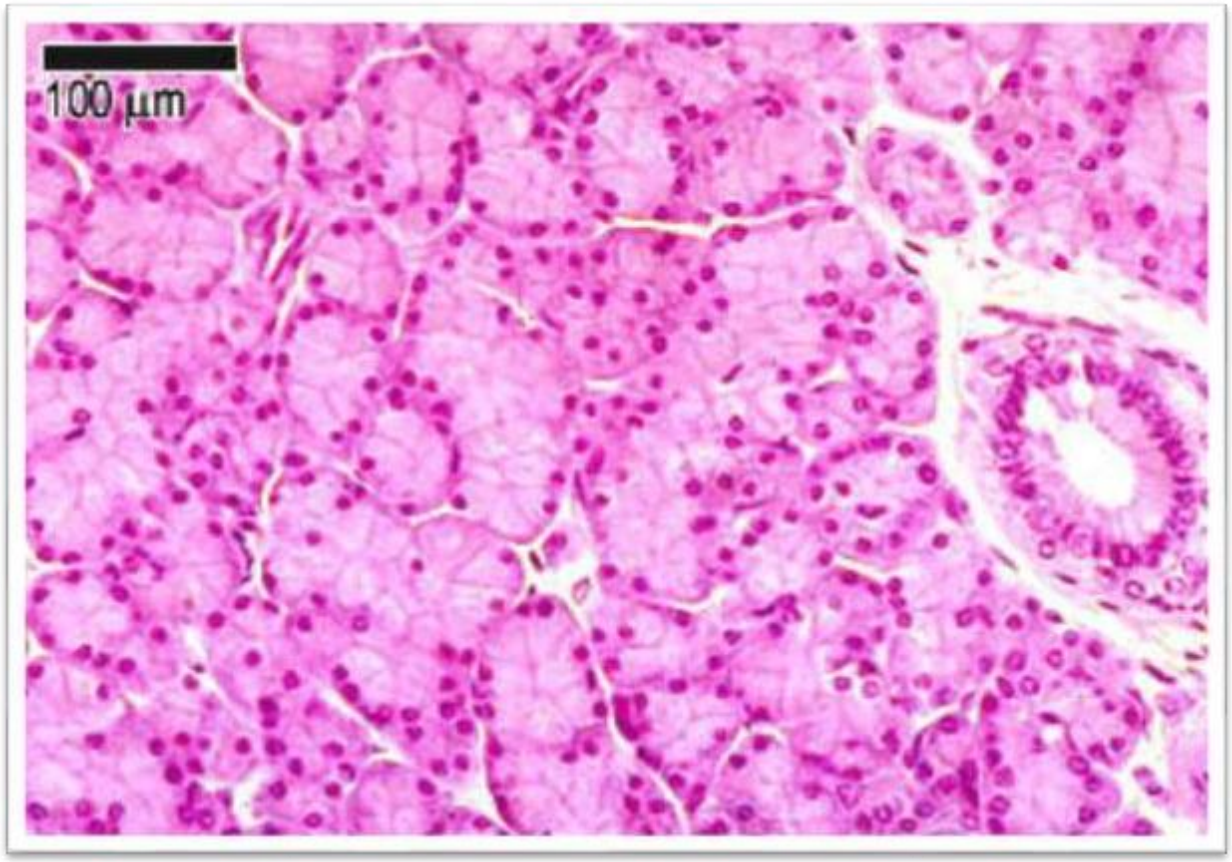


In H&E sections:

- Mucinogen granules are water-soluble

- Mucous is lost during routine tissue preparation

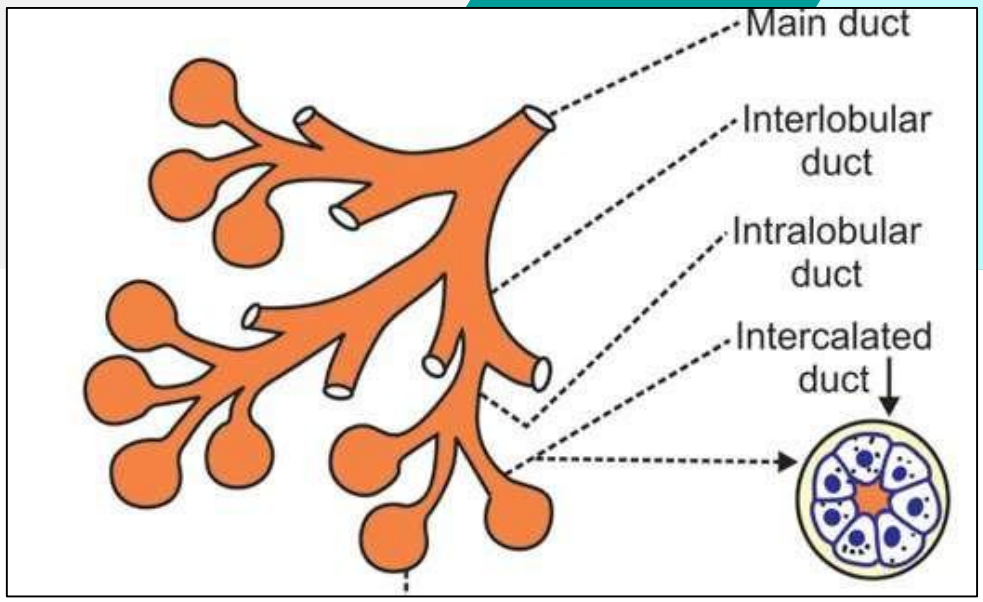
- Cytoplasm appears empty in H&E sections

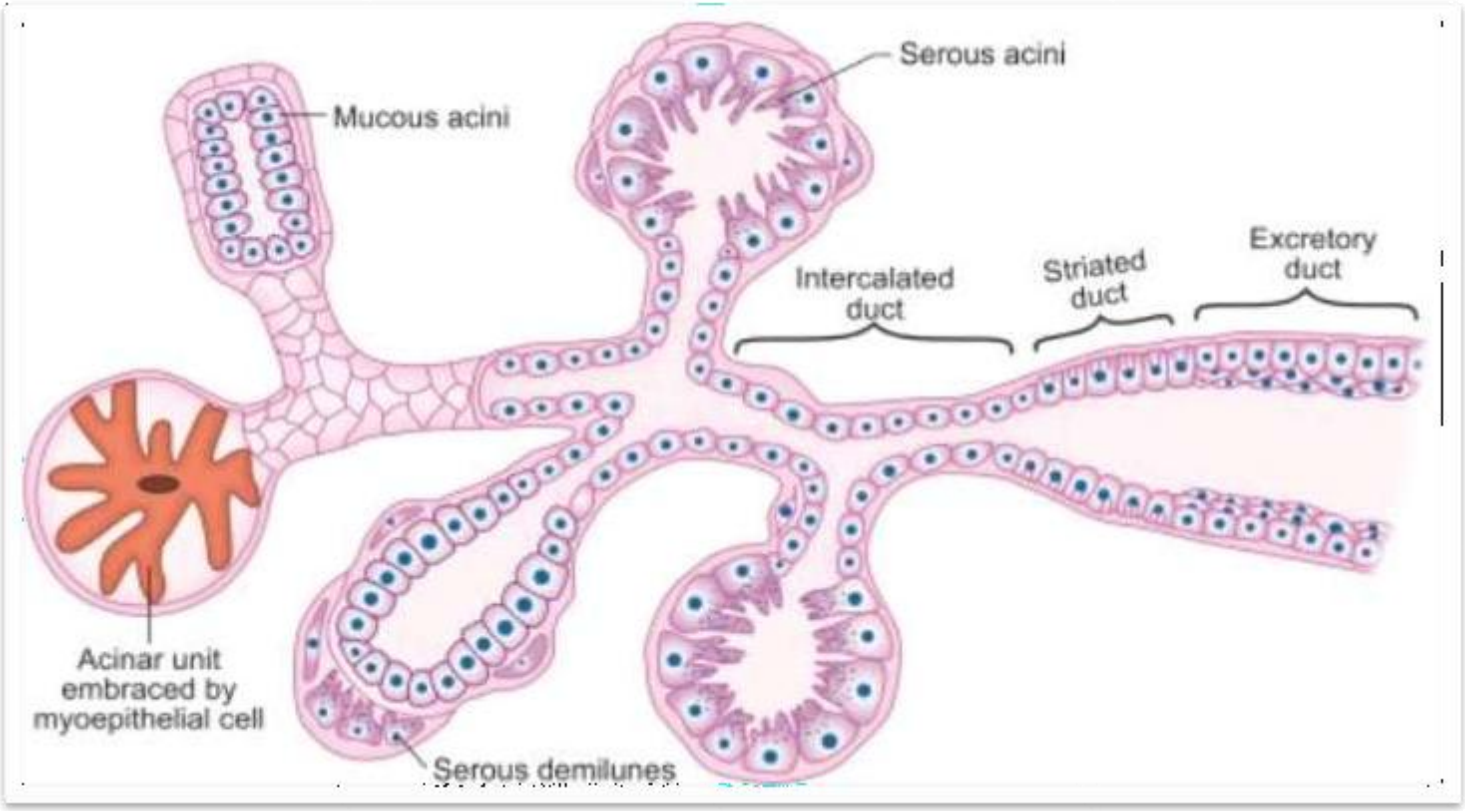


Ductal unit

The duct system has 4 segments:

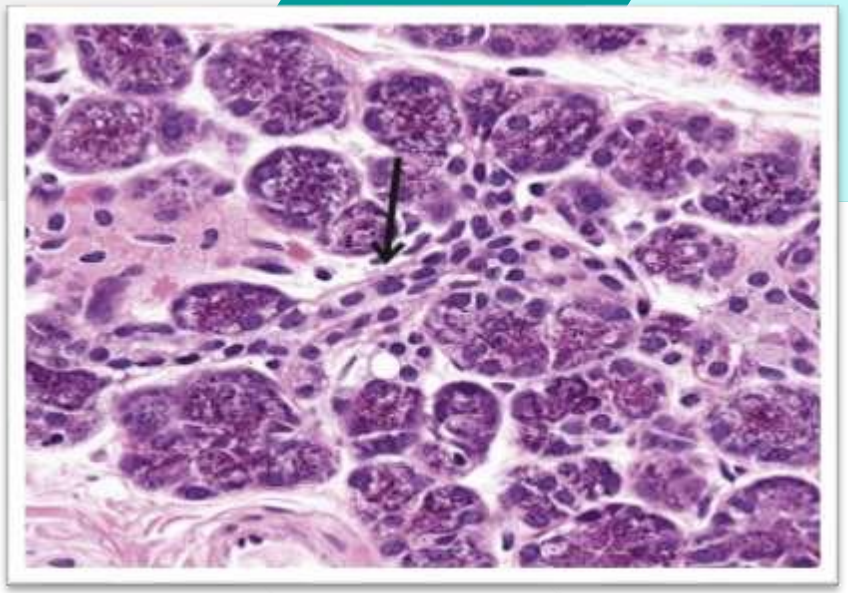
1) Intercalated duct
2) Striated duct
3) Intralobular duct
4) Interlobular duct

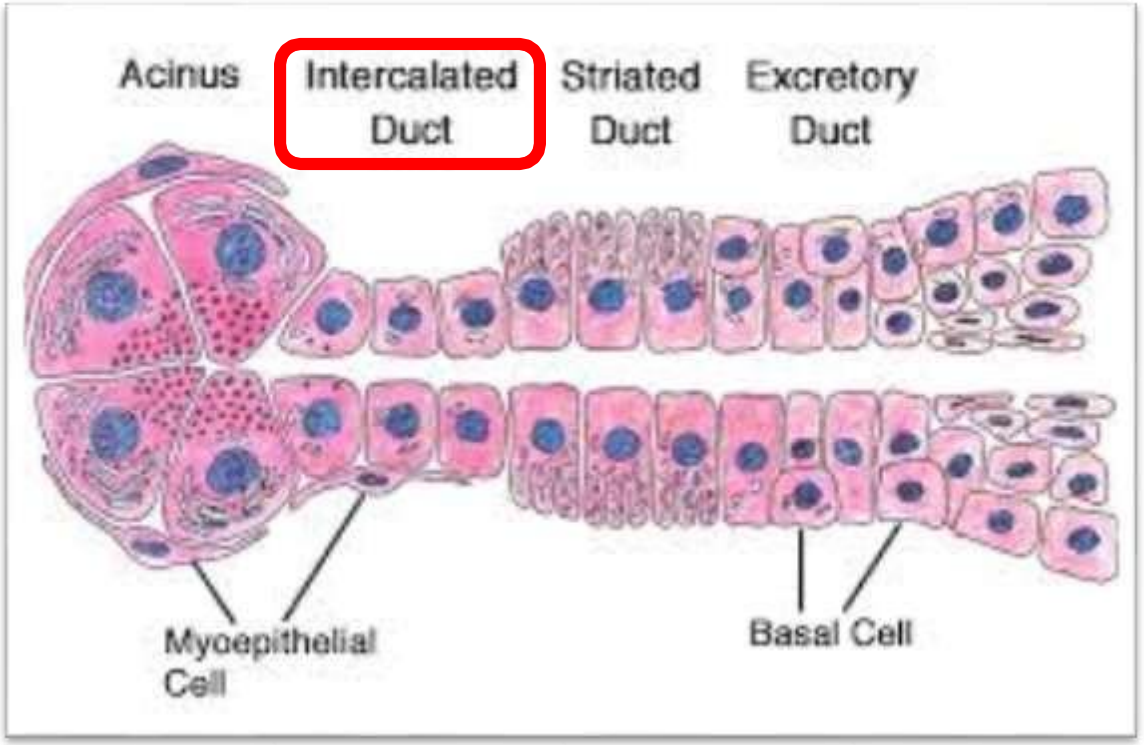




1. Intercalated duct

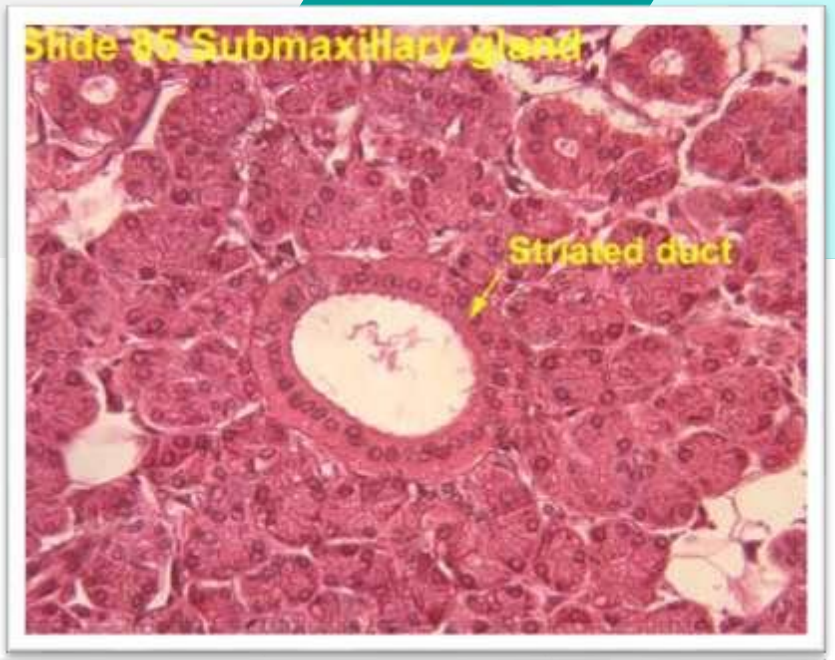
- Located between a secretory acinus & striated duct.
- Lined by low cuboidal cells.
- Duct modify the secretion in serous & mixed glands.
- Most prominently seen in **serous glands** to modify the secretion.
- Well- developed in parotid, shorter in submandibular and absent in sublingual glands.

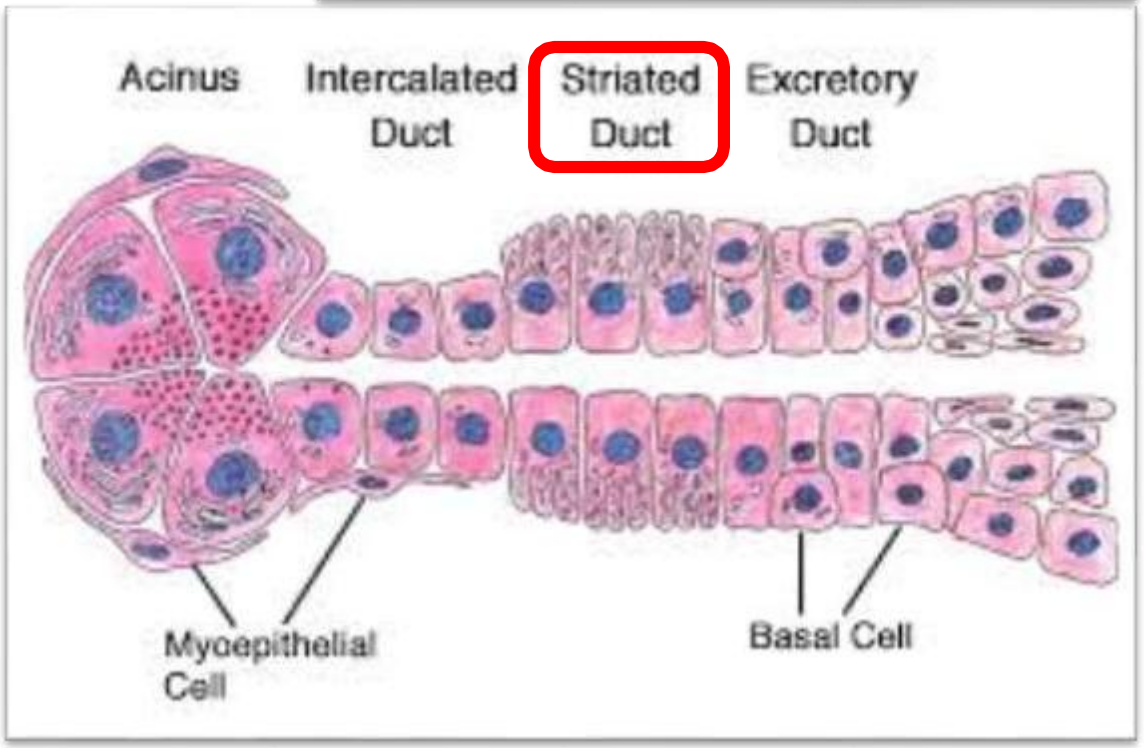




2. Striated duct

- It connects an intercalated duct to an interlobular excretory duct.
- Larger, lined by simple columnar epithelium.
- Well-developed in **serous glands.**
- Have a **basal striated appearance** due to basal folding of cell membrane.
  - ✓to enable active transport of substances out of the duct
- Saliva is modified in striated duct and produce a hypotonic fluid.



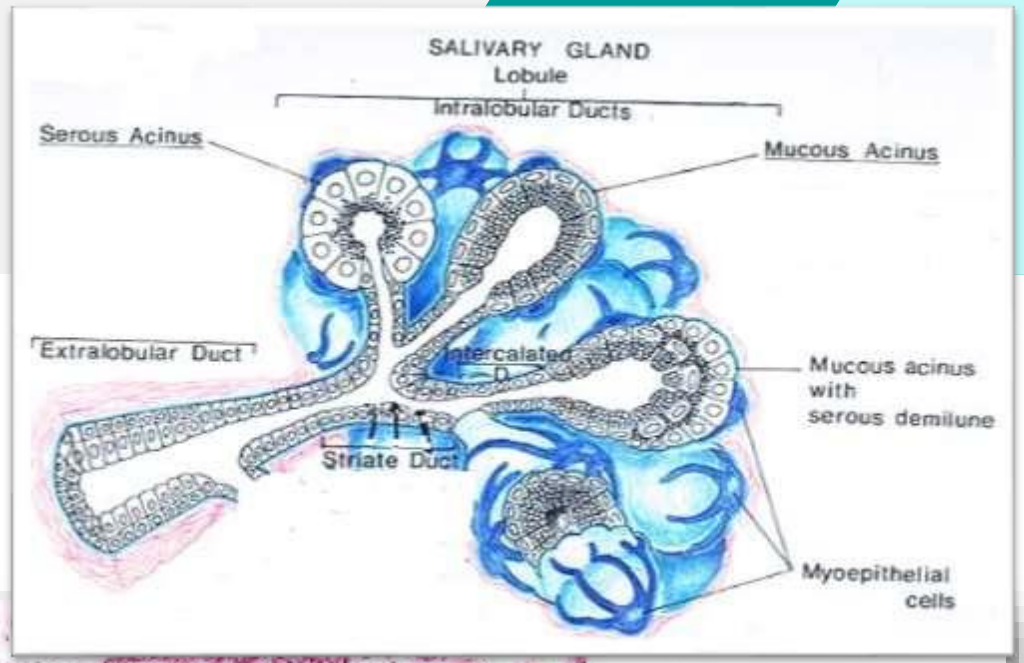


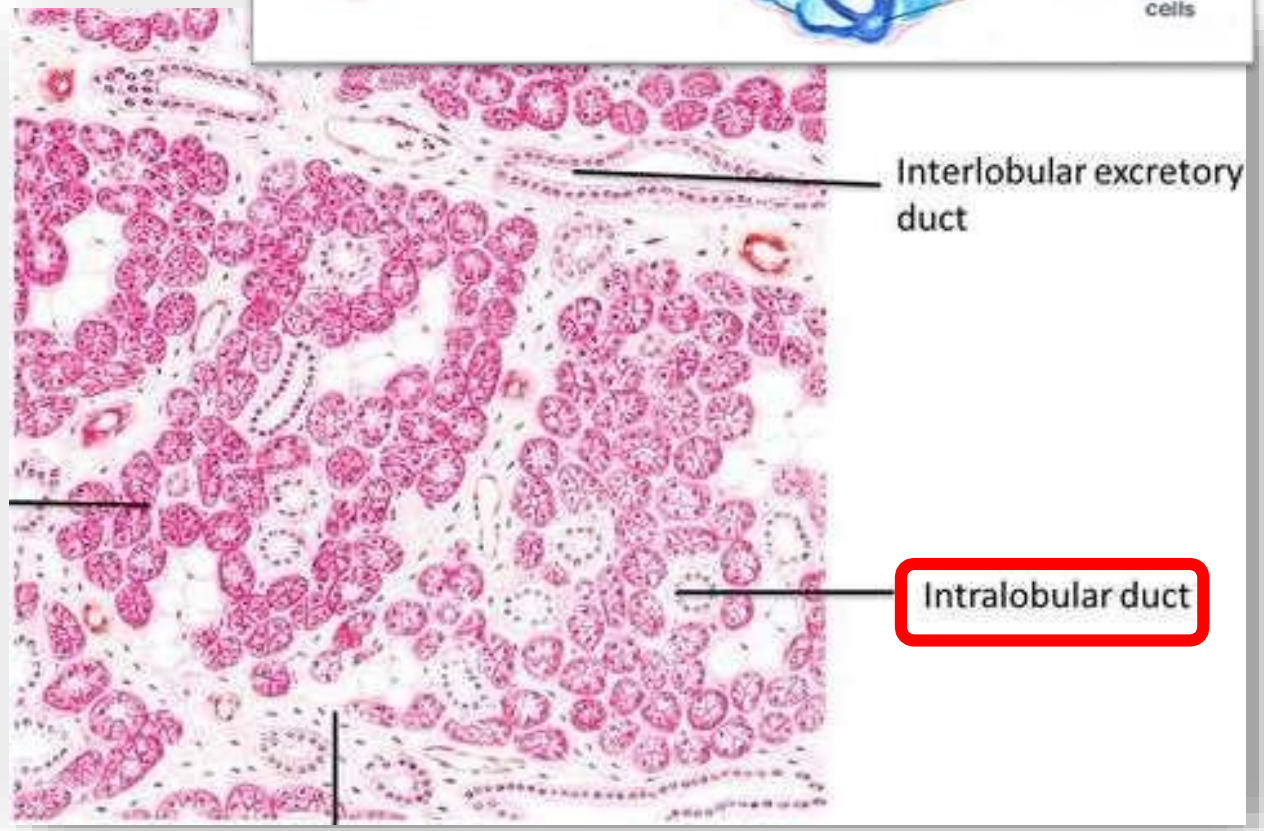
3. Intralobular duct

- An intralobular duct has two parts:
  - ✓intercalated part
  - ✓striated part

- Intralobular duct is located in the parenchyma of glands.

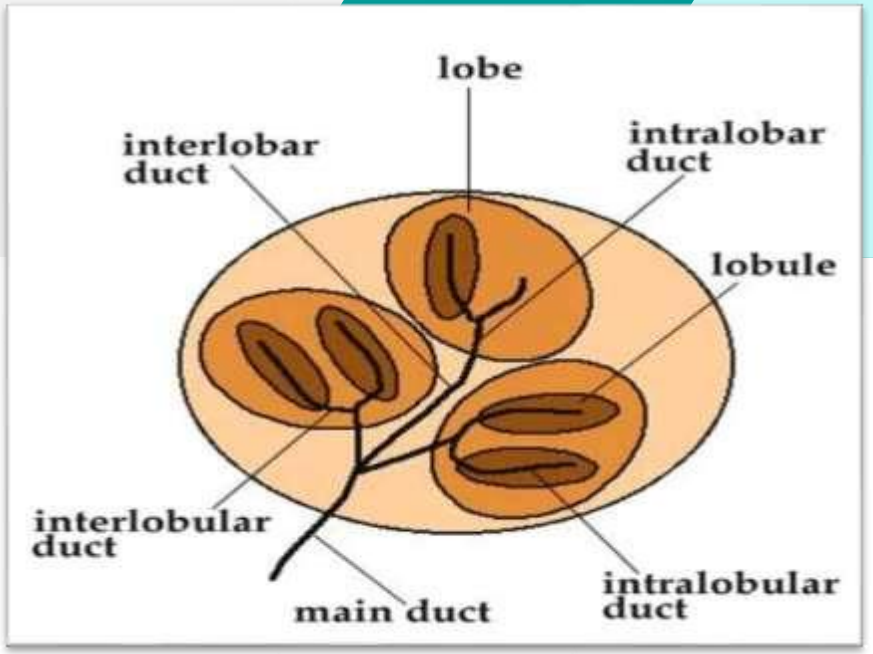
- In salivary glands, the striated ducts flow into the interlobular ducts.

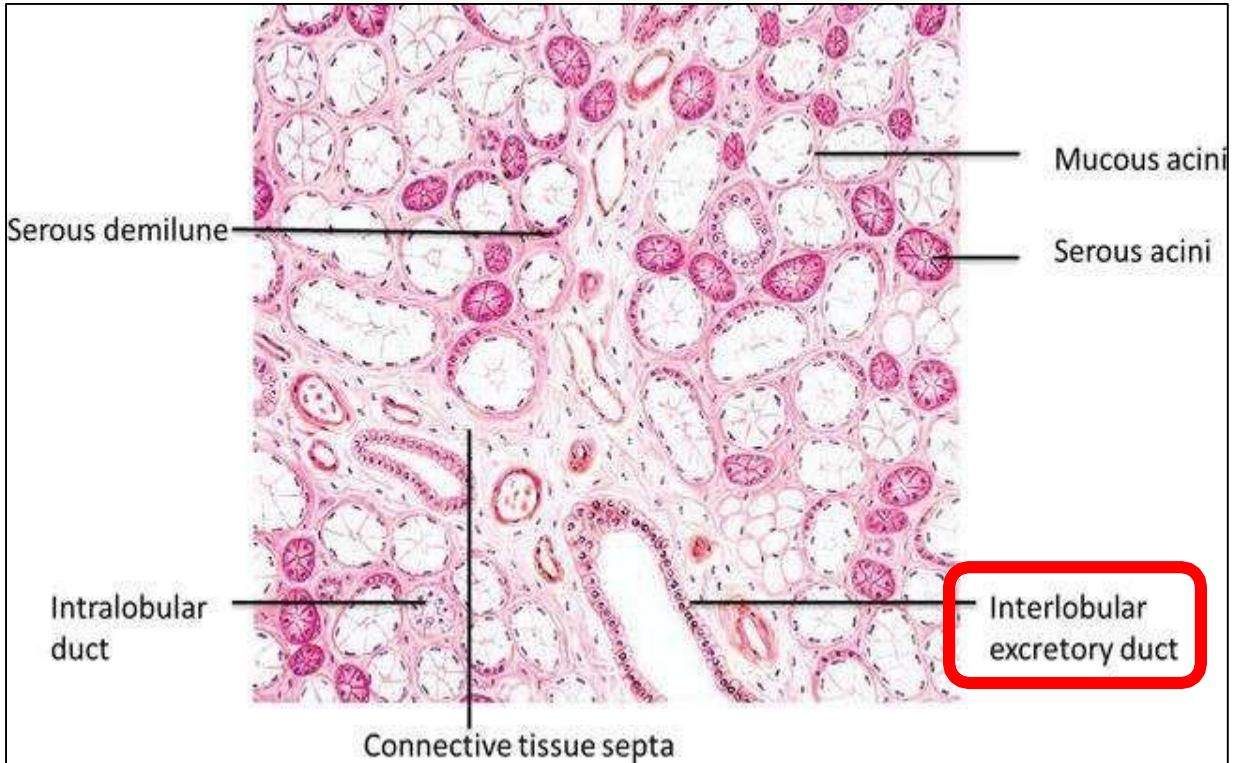




4. Interlobular duct

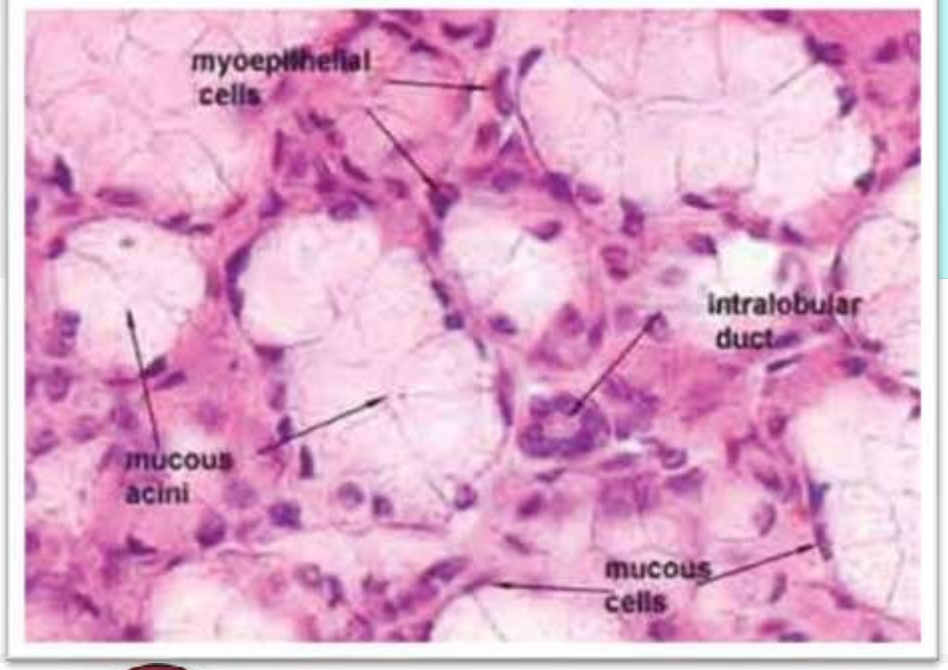
- Connect striated duct to oral cavity.
- Their lumens become progressively wider & epithelium become taller.
- Epithelium varies from columnar to pseudostratified or stratified columnar as it approaches oral epithelium.
- Epithelium with multiple layers of cells are called excretory ducts.
- Parotid & submandibular ducts travel in connective tissue before penetrating oral mucosa.

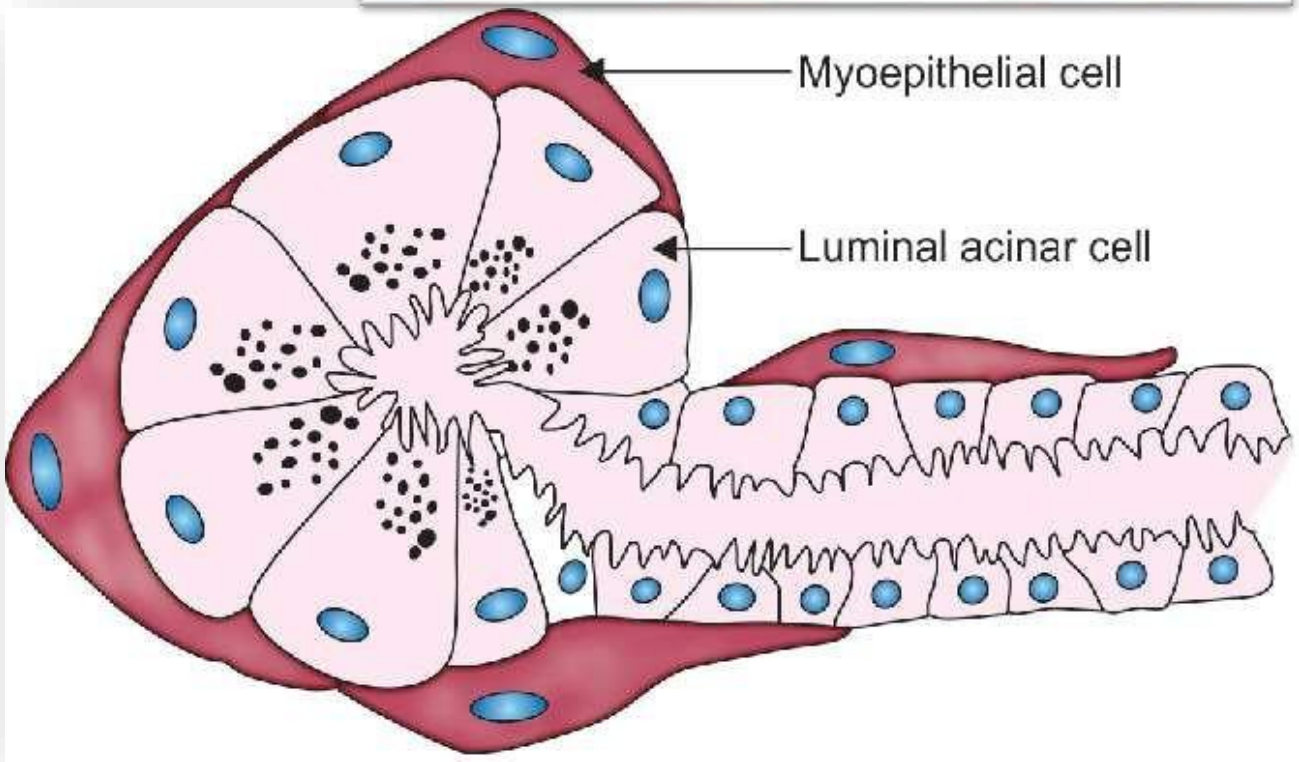




Myoepithelial cells

- Contractile cells with numerous processes.
- Lie between basal plasma membrane of epithelial cell & basal lamina of epithelium.
- Facilitates discharge of secretory products out of gland.
- In H&E section: flattened nuclei near basement membrane.





Salivary glands

There are 3 major salivary glands in the oral cavity:

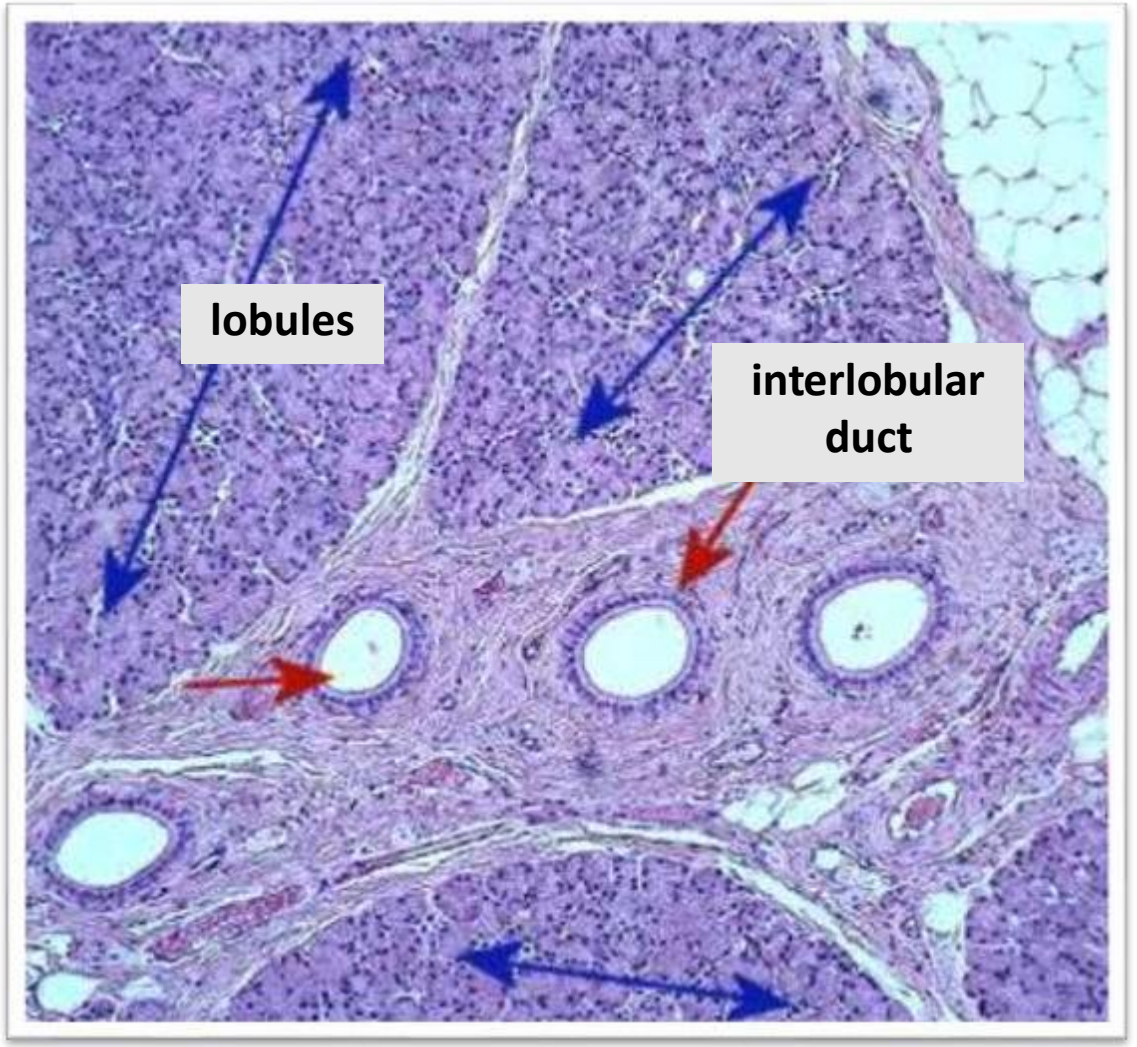
1) Parotid salivary gland

2) Submandibular salivary gland

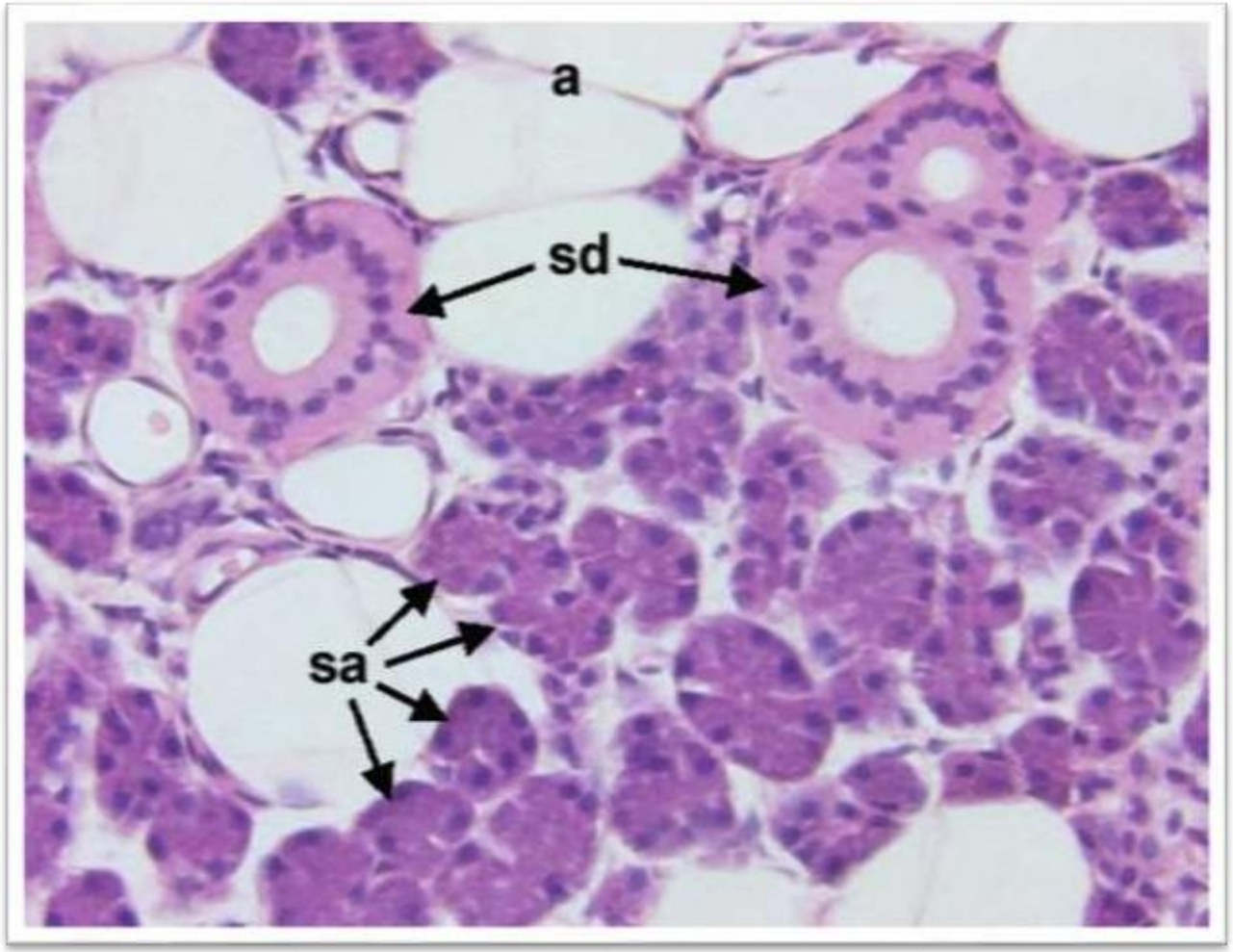
3) Sublingual salivary gland

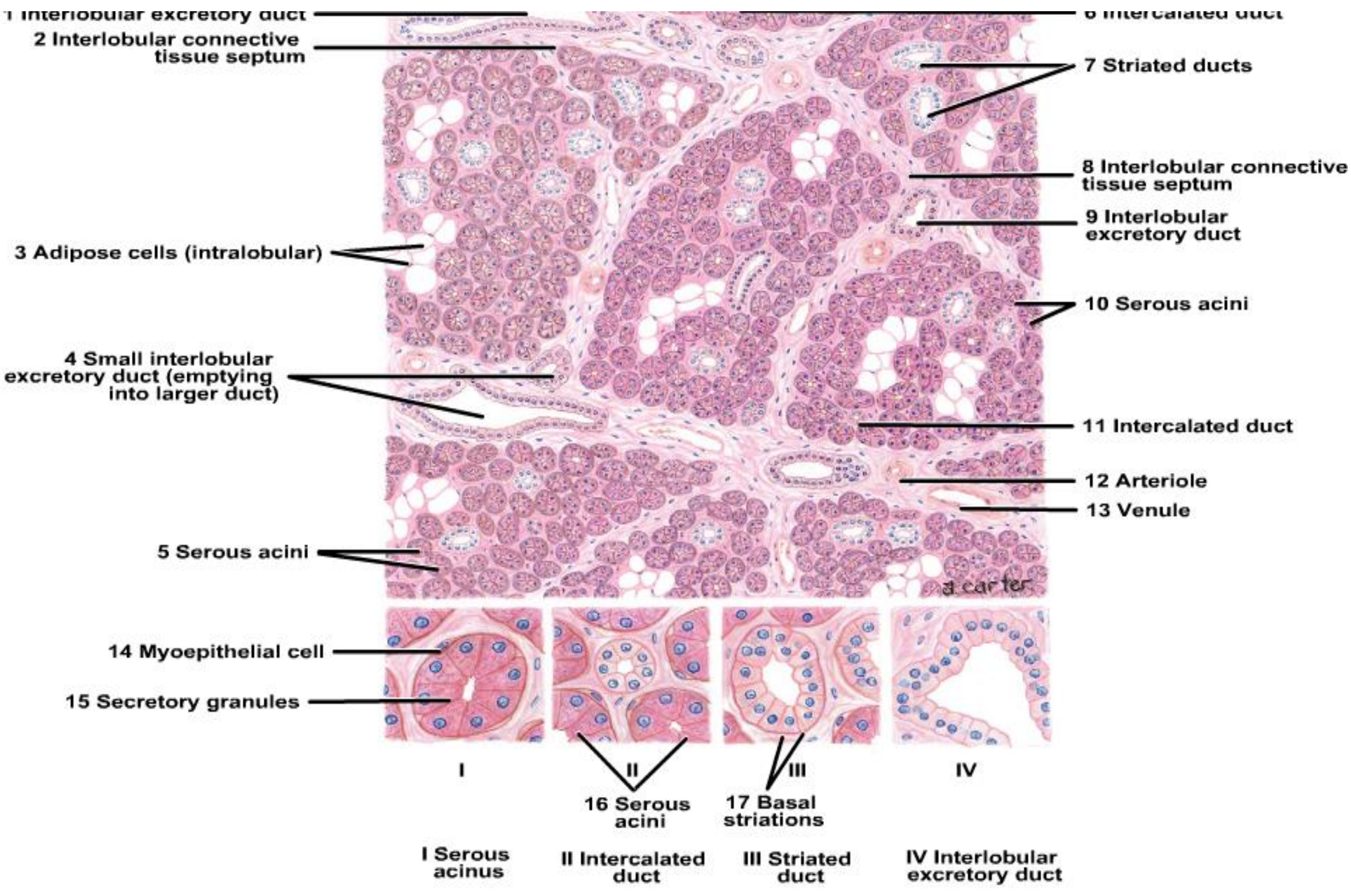
Parotid gland

- **Serous type**, compound tubuloacinar gland.
- Surrounded by a **dense connective tissue capsule**; forms septa which divides gland into lobules.
- Located in connective tissue septa are blood vessels, adipose cells, nerves & interlobular ducts.



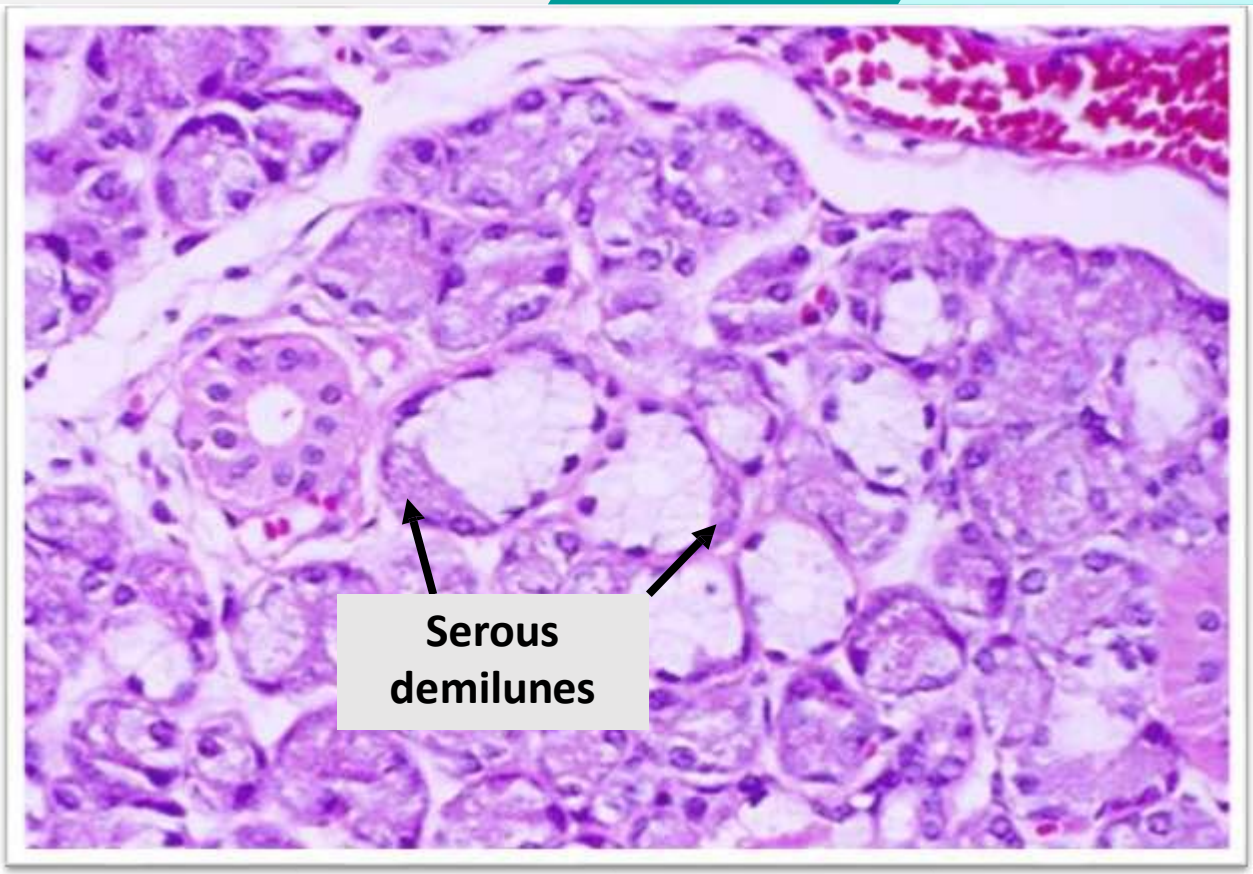
- Within each lobule are numerous serous acini **(sa)**, isolated adipose cells **(a)** and striated ducts **(sd)**.
- Abundance of **adipose cells** which increases with age.
- It contains prominent intercalated & striated ducts.



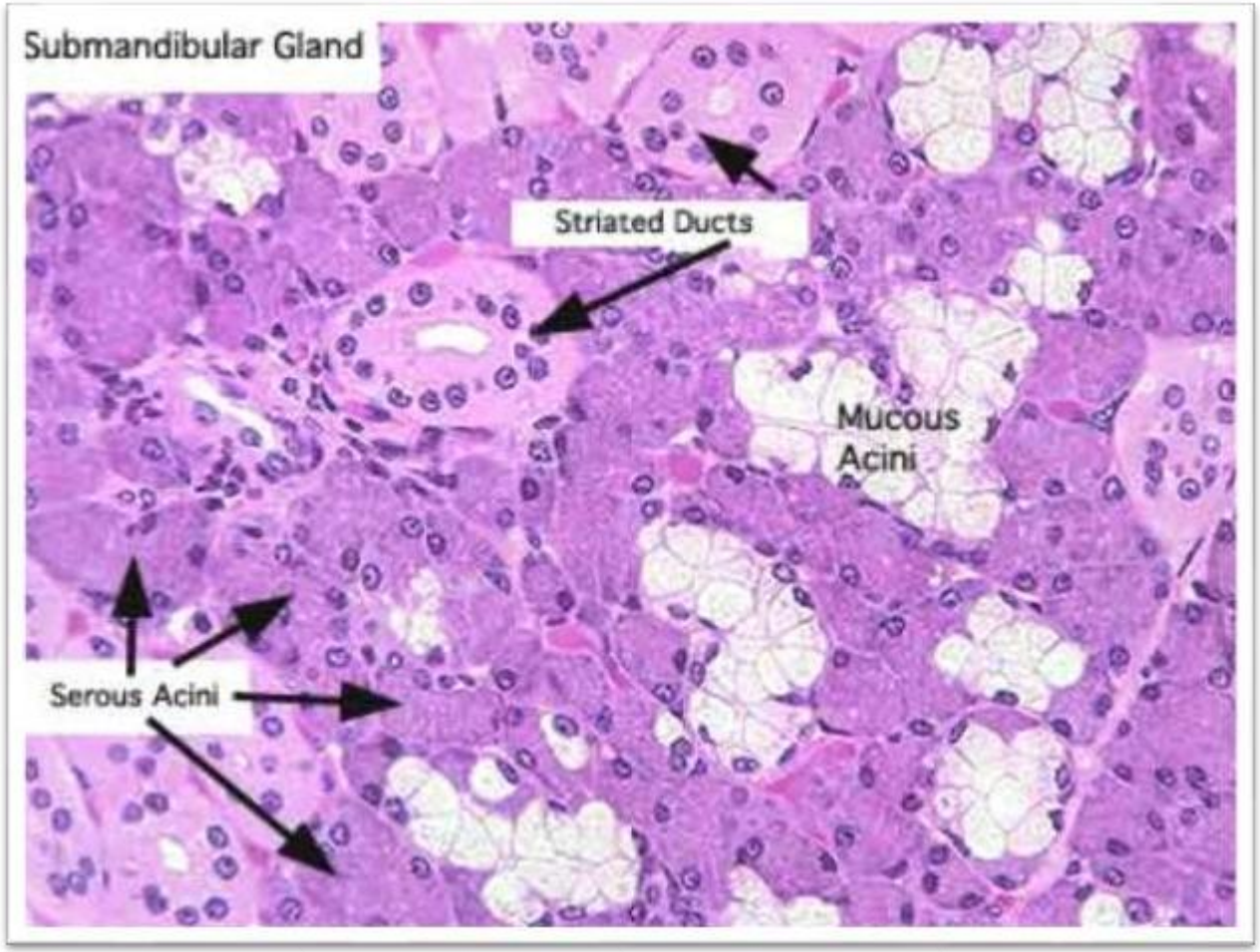

1 Interlobular excretory duct
2 Interlobular connective tissue septum
3 Adipose cells (intralobular)
4 Small interlobular excretory duct (emptying into larger duct)
5 Serous acini
6 Intercalated duct
7 Striated ducts
8 Interlobular connective tissue septum
9 Interlobular excretory duct
10 Serous acini
11 Intercalated duct
12 Arteriole
13 Venule
14 Myoepithelial cell
15 Secretory granules
16 Serous acini
17 Basal striations
I
II
III
IV
I Serous acinus
II Intercalated duct
III Striated duct
IV Interlobular excretory duct

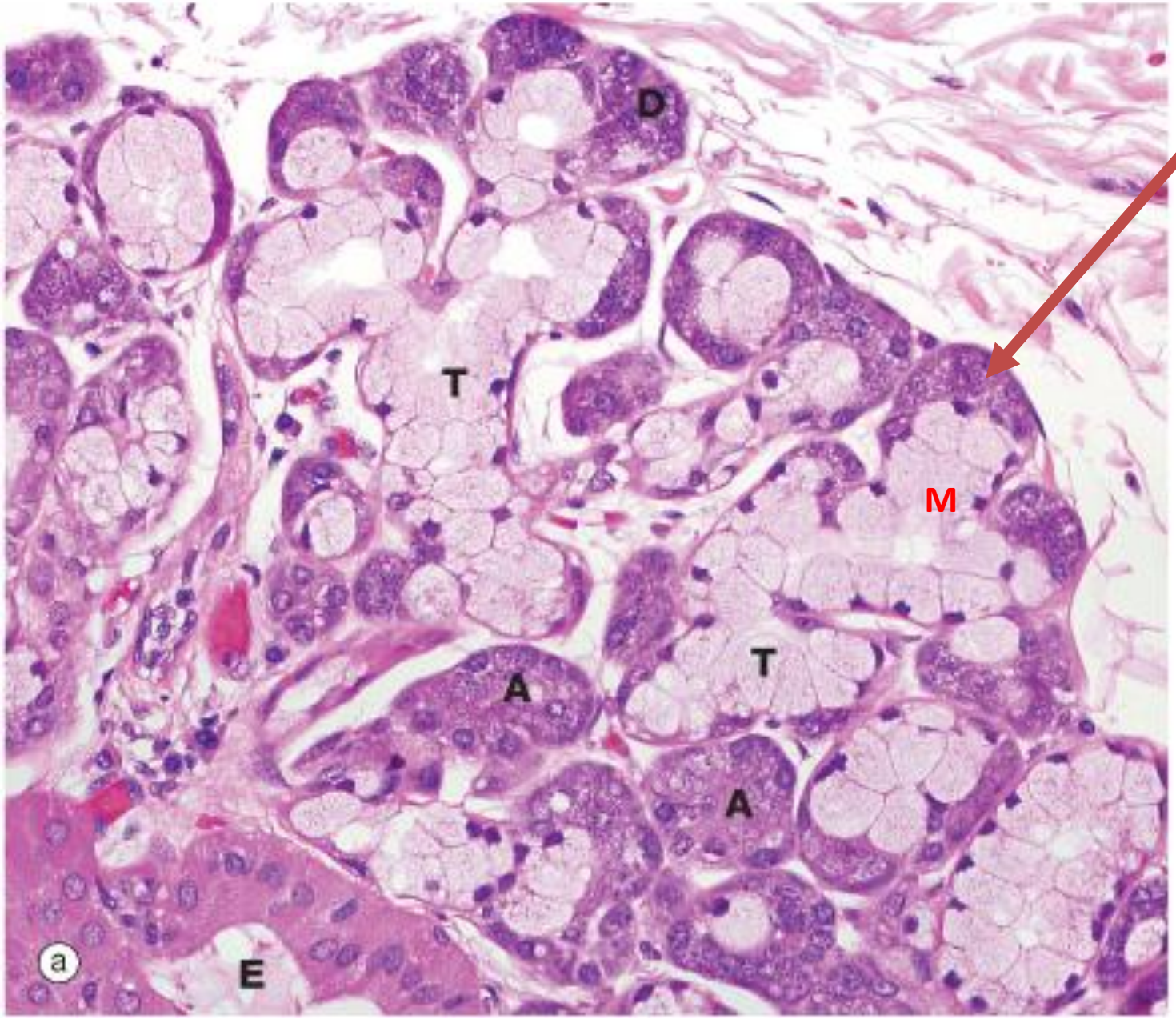
Submandibular gland

- Contain both serous and mucous acini. (**mostly serous**)
- Some mucous acini are covered with **serous demilunes**.
- Surrounded by a thin connective tissue capsule.
- Septa from capsule divide the gland into lobes.



- Mucous acini are larger.
- Exhibit pale cytoplasm with routine stain.
- Presence of intercalated and striated ducts.



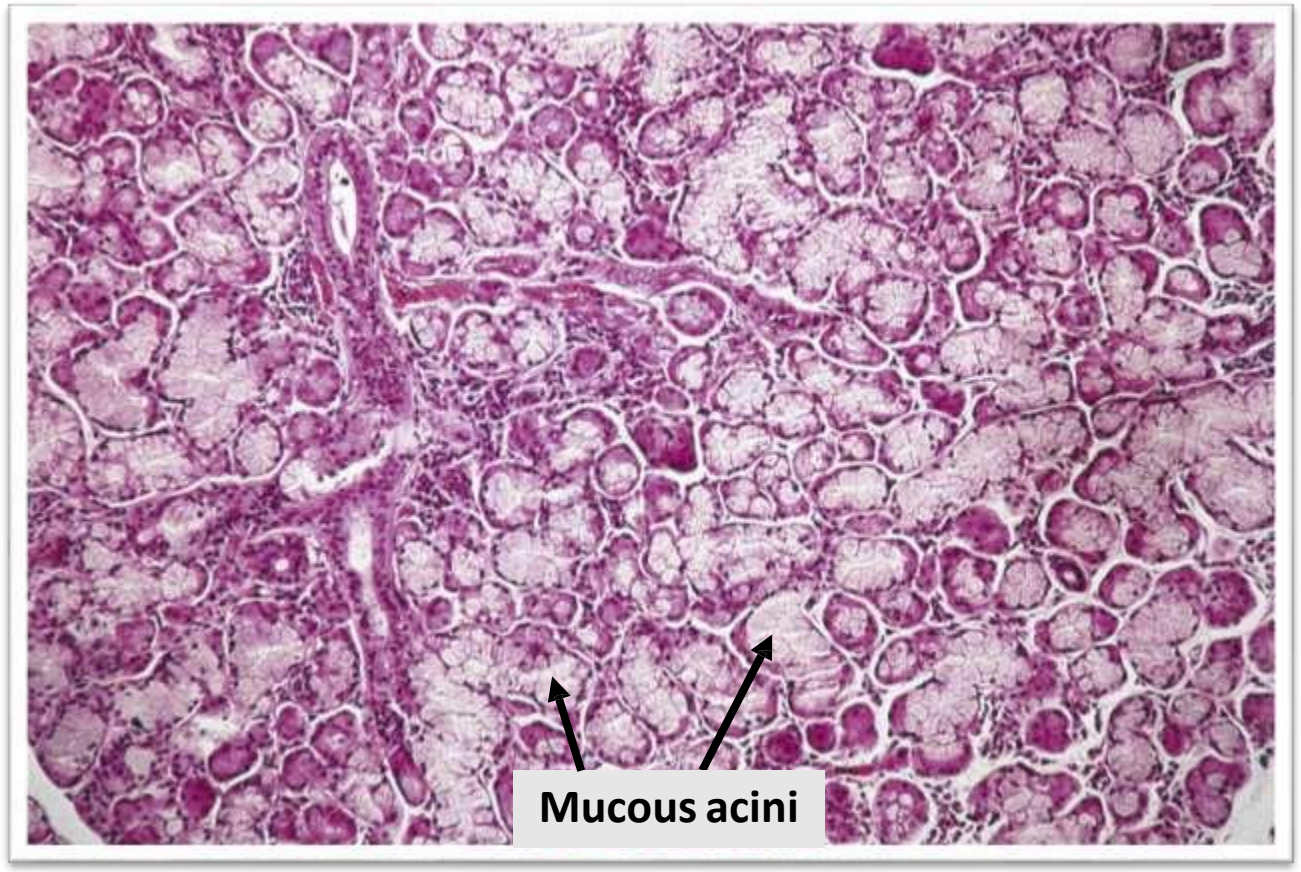


Serous demilune (semilunar shape serous Secretory unit over mucus Acinus.



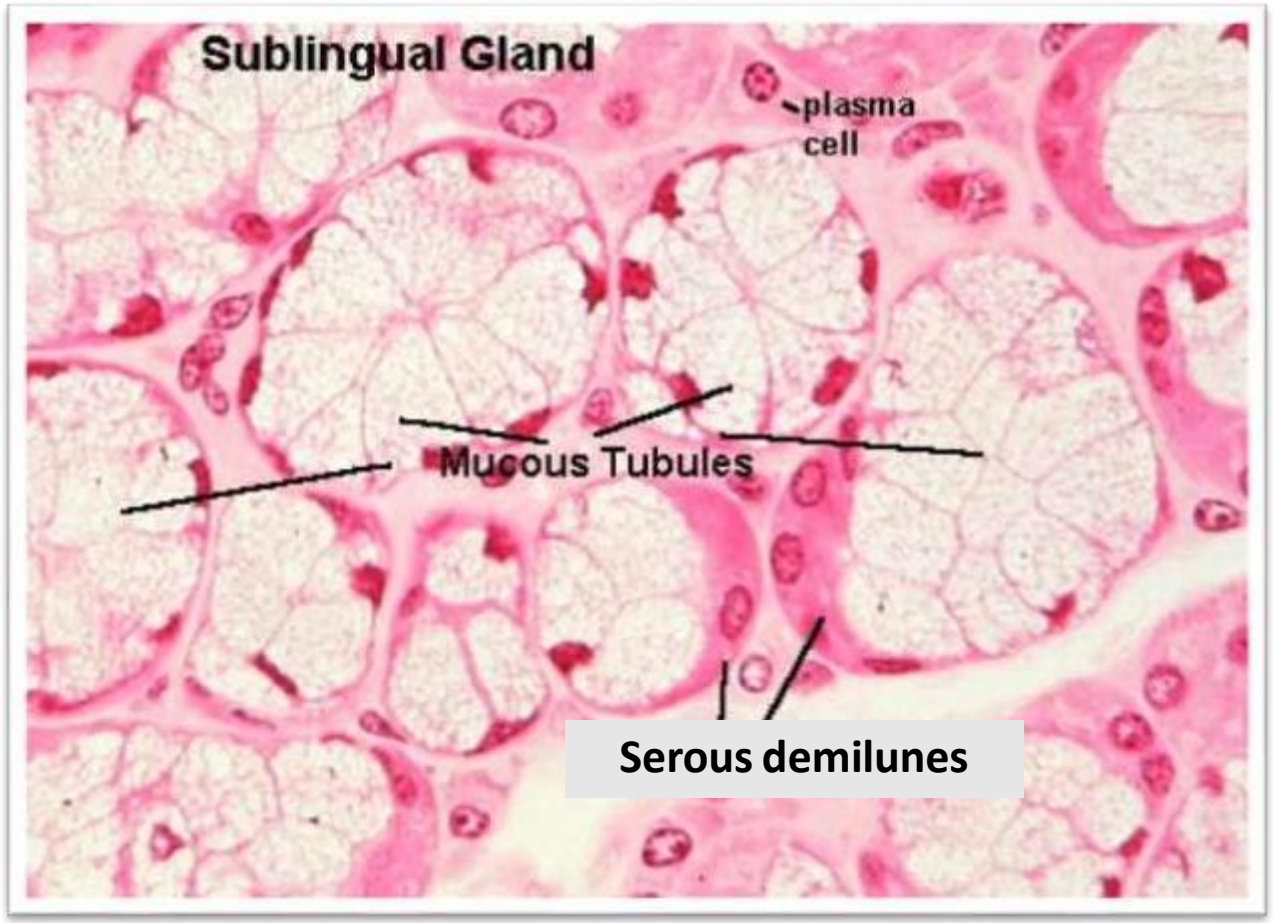
Sublingual gland

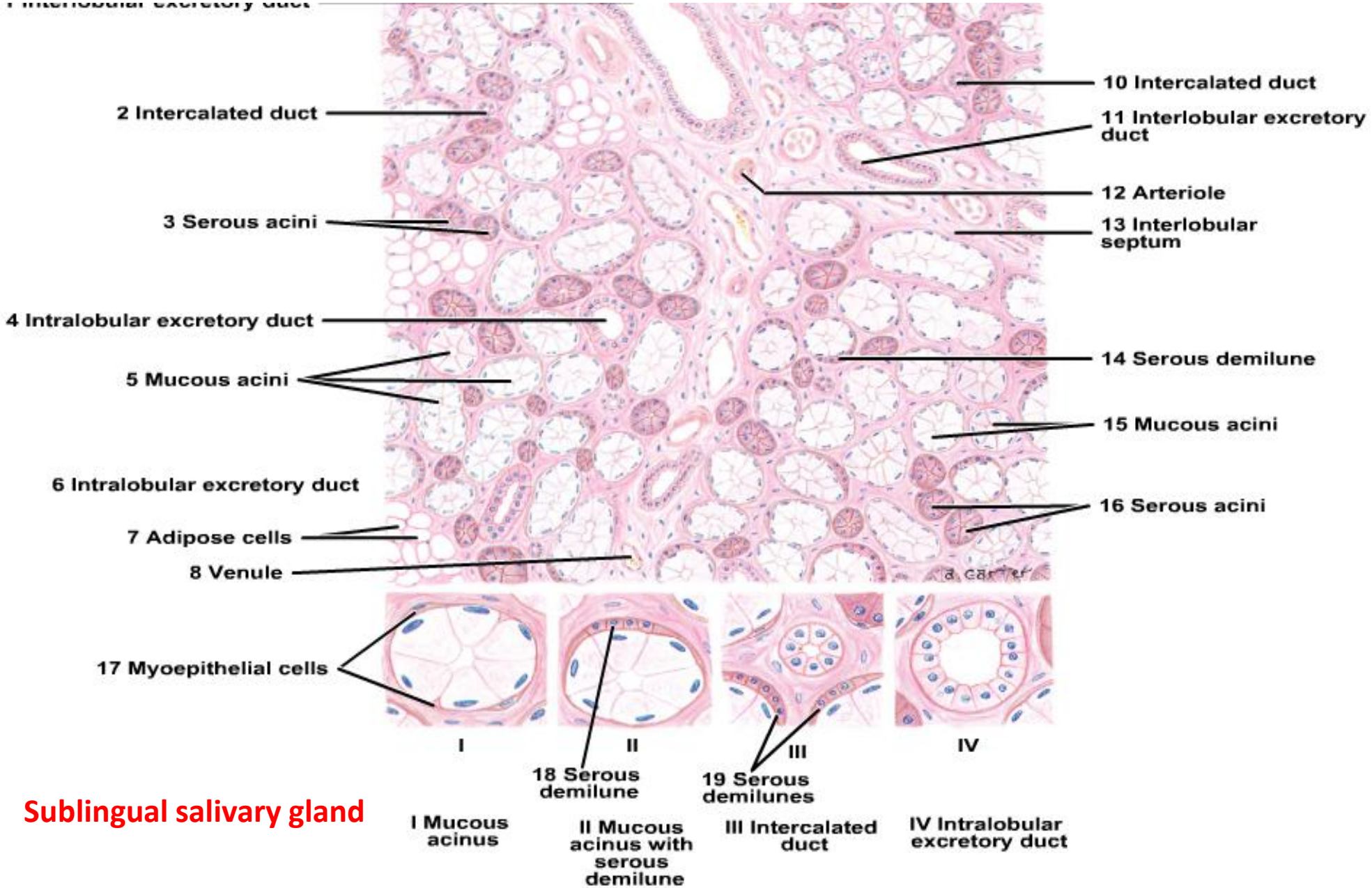
- Mixed serous and mucous gland with predominantly **mucous acini.**
- Does not contain a distinct connective tissue capsule.
- Interlobular C.T. septa are more abundant than the other two salivary glands.



- **Serous demilunes** present.

- Intercalated & striated ducts are short & difficult to locate.





Sublingual salivary gland

THANK YOU !